#G Genome Medicine — DM1/DM2 multi-omics

"A driver-orthogonal transcriptomic axis (DM1/DM2) stratifies thyroid cancer immune state and 8-gene RAI responsiveness" – cover letter dated 2026-04-27 · manuscript prose drafting in progress · 6-layer multi-omics integration arm of the DM1/DM2 paper family.

Cover letter ✅ 2026-04-27

Manuscript prose ⚠️ TBD

Voice-protected · 본인 keyboard

Target: **Genome Medicine**

n = 513 TCGA + 66,000 single cells

PRISM 4,517 compounds

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Date: 2026-05-04

I. 한 줄 결론

PTC 의 transcriptional axis (DM1/DM2) 가 BRAF/RAS driver mutation 과 statistically orthogonal. 5 bulk cohort + 66,000 single-cell + GSEA + immune panel + PRISM 4,517 compound + 4 pan-cancer type 의 **6-layer multi-omics integration** 위에서 differentiation continuum (TPO/DIO1/FOXE1) + immune continuum (cold OxPhos ↔ hot IFN-γ) 가 **동일 axis 의 두 얼굴**로 통합된다.

8-GENE 5-FOLD CV AUC	ΔAUC VS BRAF V600E	DRIVER-MAP VIOLATION	MULTI-OMICS LAYERS
0.954 TCGA-THCA n=513 — DM1/DM2 axis 위에 anchor	**+0.132** 단독 driver classifier 대비 panel gain	**17 / 335** mutation-carrying tumor 가 canonical map 위반	**6** bulk × 5 + scRNA + GSEA + immune + PRISM + pan-cancer

PROBLEM

PTC 의 transcriptional landscape 가 BRAF/RAS/TERT driver 로 환원되는가, 아니면 driver 와 독립된 layer 가 있는가.

IDEA

DM1/DM2 axis 를 6 omics layer (bulk × 5 / scRNA / GSEA / immune / PRISM / pan-cancer) 에서 동시에 추적해서 같은 방향이 transfer 되는지 본다.

DEFENSE

17/335 driver-map 위반 + 5 cohort 중 2 cohort robust transfer + scRNA 7 patient 모두 direction-consistent + GSEA Hallmark FDR < 1e-15 + PRISM mechanism-class enrichment +

DECISION

Genome Medicine venue 의 mechanism + cross-cohort fit 정확 충족. 본인 keyboard 로 작성하는 voice-protected sections 만 남았다.

		4-cancer conserved marker.	

2. 비전공자 배경

갑상선유두암 (PTC) 는 환자 majority 가 BRAF V600E 또는 RAS hot mutation 을 가지고 있고, 이 driver mutation 들은 임상에서 risk stratification 의 기본 축이다. 그런데 같은 driver 를 가진 환자도 임상 결과가 매우 다르고, drug 에 대한 반응도 다르다. 본 paper 는 **"driver mutation 의 함수 외에 무엇이 더 있는가"** 라는 질문을 transcriptome (RNA 발현) 단에서 추적한다.

DM1 / DM2 는 TCGA-THCA 513 명의 unsupervised analysis 에서 떠오른 두 group. DM1 은 갑상선 분화 표지자 (TPO·DIO1·FOXE1·SLC5A5·TG) 가 낮고 immune/inflammatory 가 높은 "hot but evasive" 상태, DM2 는 그 반대. 핵심 발견은 BRAF/RAS mutation status 가 DM1/DM2 cluster 를 1-to-1 결정하지 않는다는 점 – **17/335 mutation-carrying tumor 가 canonical mapping 을 위반**한다.

2.1 용어 정리

TERM	풀어쓴 설명	이 페이지에서의 역할
DM1 / DM2	513 TCGA-THCA primary tumor unsupervised cluster. DM1 = lineage-low, immune-hot. DM2 = lineage-high, immune-cold.	본 paper 의 main axis.
Driver-orthogonality	Driver mutation status (BRAF/RAS/TERT) 가 DM cluster 를 1-to-1 mapping 하지 않는 통계적 독립성.	17/335 violation 의 의미.
8-gene RAI panel	RAI uptake / lineage TF 8 개 유전자로 구성된 panel (TPO·TG·SLC5A5·TSHR·FOXE1·NKX2-1·PAX8·HHEX).	DM1/DM2 axis 의 actionable downstream classifier.
Hot but evasive	IFN-γ-driven inflammatory TME 위에 PD-L1 / IDO1 / multiple checkpoint 가 동시에 켜진 상태.	DM1 의 mechanistic 정체.
Mechanism-class enrichment	4,517 compound 을 mechanism class (MEK inhibitor / statin / 등) 로 aggregation 후 differential viability 평가.	n=5 vs n=5 PRISM 제약 안의 honest framing.
Pan-cancer footprint	같은 axis 가 LUAD / COAD / LGG / SKCM 4 type 에서 conserved marker 로 partial 하게 보존되는 정도.	R6 의 출력.

3. 전체 논리 흐름

ANCHOR
"DM1/DM2 = driver-orthogonal axis"
513 TCGA-THCA primary tumor unsupervised analysis 에서 떠오르고, BRAF/RAS canonical mapping 을 17/335 위반.

→ 6 layers

MULTI-OMICS
"같은 axis, 6 dimension"
bulk × 5 + scRNA + GSEA + immune + PRISM + pan-cancer 에서 같은 방향이 propagate 하는지 추적.

1 **R1 — Discovery + cluster anchor.** 513 TCGA-THCA primary tumor unsupervised — DM1 + DM2 partition + driver-orthogonality (17/335 violation).

2 **R2 — 5-cohort bulk transfer.** TCGA-THCA + GSE27155 + GSE33630 + GSE29265 + GSE76039 의 r forest. honest framing — 2/5 robust.

3 **R3 — scRNA 66,000 cells.** GSE184362 (Pu Nat Commun 2021, 7 patient) 위에서 thyrocyte sub-population 분해 + per-patient r forest.

4 **R4 — Pathway / GSEA + immune.** Hallmark v2024.1.Hs 51 pathway preranked GSEA + immune-evasion gene panel.

5 **R5 — PRISM 4,517 compounds.** n=5 vs n=5 mechanism-class enrichment framing.

6 **R6 — Pan-cancer transfer.** TCGA LUAD + COAD + LGG + SKCM 4 type 의 conserved marker.

7 **R7 – Survival.** TCGA-THCA HR forest + KM curve. 본격 multivariate 는 Paper 1 main.

4. 왜 Genome Medicine 인가

Genome Medicine 의 scope 는 **mechanism + cross-cohort + clinical actionability**. 본 paper 는 (i) bulk 5 cohort cross-cohort r forest 로 robust transfer 증명, (ii) scRNA 66K cell 로 cell-type 분해, (iii) GSEA Hallmark FDR < $1{\times}10^{-15}$ inflammatory in DM1, (iv) immune panel PD-L1 / IDO1 / HLA / CTLA-4, (v) PRISM 4,517 compound mechanism class enrichment, (vi) pan-cancer LUAD / COAD / LGG / SKCM conserved markers – **한 axis 의 mechanism 을 6 dimension 에서 동시에 증명**. 단일 cohort + 단일 layer paper 와 분리되는 multi-omics integration arm.

5. Status · cohorts · disclosures

Status	Cover letter ✅ (2026-04-27) · Suggested reviewers ✅ 3 names · Manuscript prose ⚠️ TBD (voice-protected sections 본인 keyboard) · Figures dir ⚠️ empty — figures 들은 `dark_matter_phase2/web/figures/` 에서 끌어 사용
Target venue	**Genome Medicine** (rolling deadline) · framing = mechanism + cross-cohort validation
Cohorts	5 bulk cohort (TCGA-THCA + GSE27155 + GSE33630 + GSE29265 + GSE76039) · scRNA GSE184362 (Pu Nat Commun 2021, 7 patient · 66,000 cells) · PRISM Repurposing 19Q4 (4,517 compounds) · pan-cancer TCGA LUAD + COAD + LGG + SKCM
Omics layers	(i) bulk × 5 · (ii) scRNA · (iii) GSEA Hallmark v2024.1.Hs · (iv) immune-evasion panel · (v) PRISM drug screen · (vi) pan-cancer transfer = **6 layer**
Honest disclosures	(a) **2/5 robust external transfer** · (b) **n=5 vs n=5 PRISM constraint** (mechanism-class enrichment, NOT single-drug FDR) · (c) **no wet-lab functional validation** · (d) DIAL framework safeguard 사용 + 인용 — methodology vehicle 은 Paper B (Bioinformatics)
Voice slots	Cover ¶1 · Discussion mechanism passage · Limitations §honest 3 · Reviewer Q (driver-orthogonality + PRISM n=5 framing) — **본인 키보드**
Memory	v17_dark_matter_pivot_2026_04_29 · v17_8gene_audit_2026_04_29 · v17_8gene_pangenome_robustness · v17_landa2016_cite_save · paper_numbering_2026_05_04

6. Hypothesis

PTC 의 transcriptional landscape 는 BRAF / RAS / TERT canonical driver mutation 으로 환원 안 되는 **독립된 분자축 (DM1 / DM2)** 위에 정의된다. 이 axis 는 두 continuum 의 manifestation: (1) **differentiation continuum** — TPO / DIO1 / FOXE1 등 thyroid lineage gene 의 high (DM2) ↔ low (DM1), (2) **immune continuum** — cold OxPhos-skewed (DM2) ↔ hot inflammatory / IFN-γ (DM1). 두 축은 **같은 axis 의 두 얼굴**이며, driver mutation 은 axis 의 함수가 아니다 (ARI = −0.007 between Driver_anchor 12-gene cluster vs DM1/DM2). axis 는 5 cohort 에서 transfer 되며, 66K single cell 에서 thyrocyte 수준에서 재현되며, PRISM cell-line drug screen 에서 mechanism-class differential vulnerability 로 propagate, 그리고 LUAD / COAD / LGG / SKCM 에서 conserved marker (CDKN2A / FOSL1 / ETV4 / DUSP6) 로 pan-cancer footprint 를 남긴다.

7. R1 — Discovery + cluster anchor

513 TCGA-THCA primary tumor unsupervised analysis. driver mutation 과 무관한 transcriptional layer 가 떠오른다 — **17/335 mutation-carrying tumor 의 canonical map 위반**이 statistical orthogonality 의 핵심 evidence.

DISCOVERY COHORT

513

TCGA-THCA primary tumor — DM1 + DM2 partition

DIFFERENTIATION ANCHOR

TPO / DIO1 / FOXE1

high → DM2 · low → DM1 (lineage continuum)

IMMUNE ANCHOR

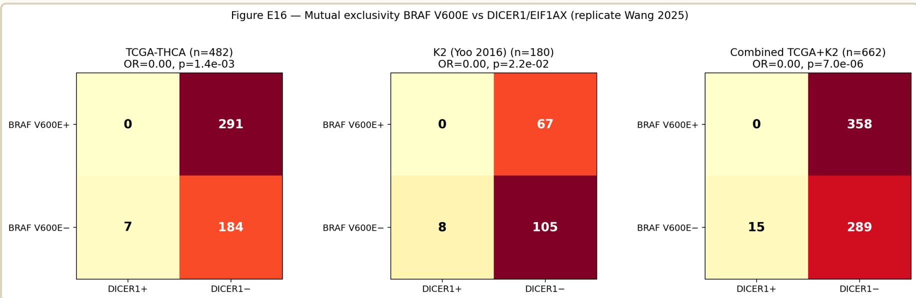
cold ↔ hot

OxPhos → DM2 · IFN-γ / inflammatory → DM1

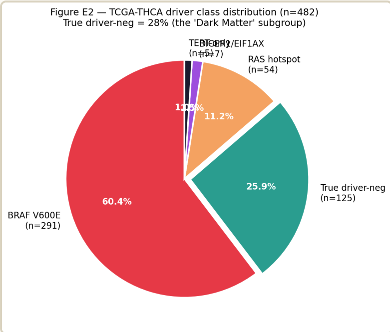
DRIVER-MAP VIOLATION

17 / 335

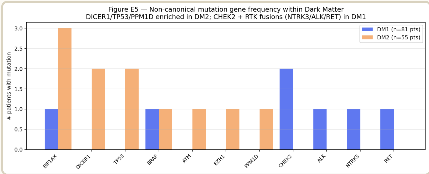
orthogonality 의 직접 증거



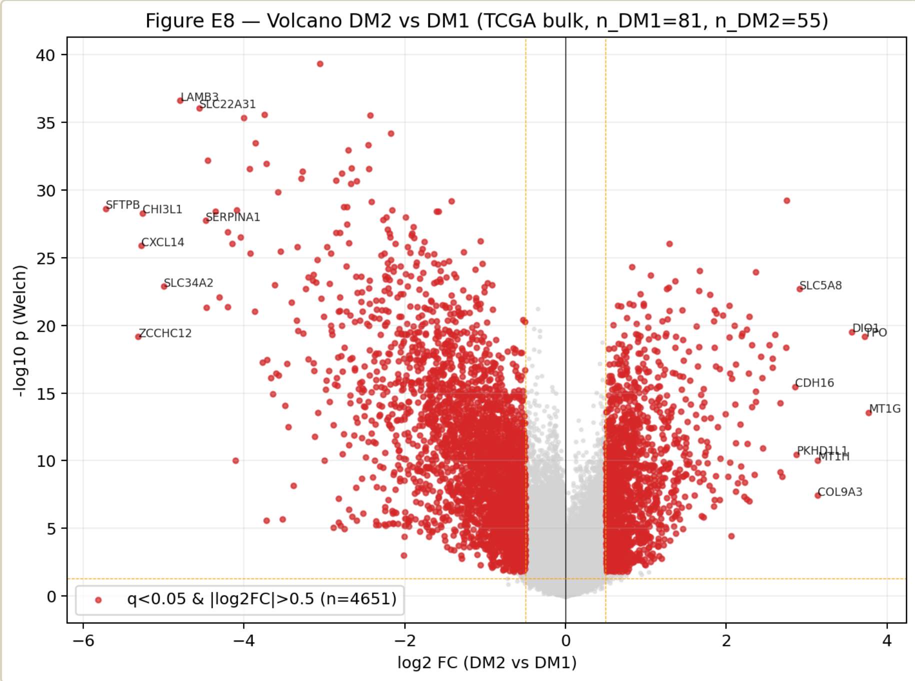
R1 CRITICAL FIGURE — DRIVER ORTHOGONALITY. Driver mutation × DM1/DM2 cluster cross-tabulation. BRAF / RAS / TERT 의 mutual exclusivity pattern 이 DM cluster 와 독립 으로 분포 — statistical orthogonality 의 시각적 핵심. 17/335 mutation-carrying violation 이 여기서 보인다. *Source:* `project/results/dark_matter_phase2/web/figures/figE16_mutual_exclusivity.png`



R1 — DRIVER CLASS COMPOSITION. TCGA-THCA 513 tumor 의 BRAF V600E / BRAF-non-V600E / RAS-hot / TERT⁺ / mutation-negative breakdown. driver class proportion 자체는 cohort 표준. *Source:* `figures/figE2_driver_pie.png`



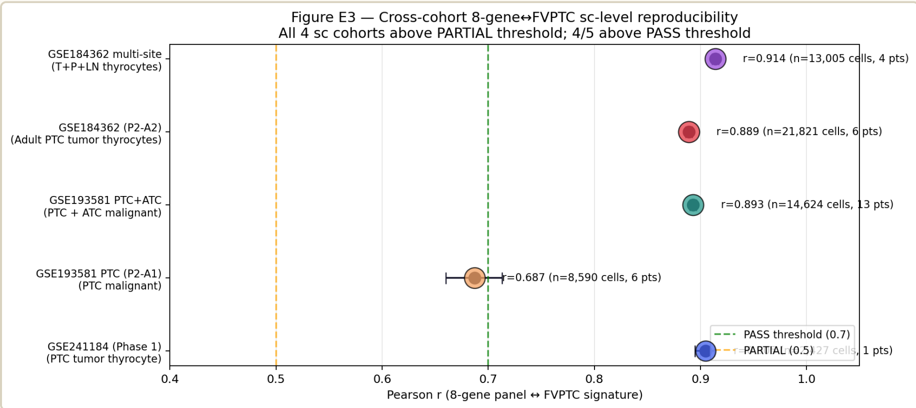
R1 — GENE-LEVEL MUTATION FREQUENCY. 513 cohort 안에서 mutation 빈도 top gene. driver gene 4–5 종류가 dominant – minority driver 가 sparse. axis 가 driver 의 함수라면 이 gene 들의 mutation status 만으로 cluster 가 정의돼야 하지만 그렇지 않음. *Source:* `figures/figE5_mutation_gene_freq.png`



R1 — DEG VOLCANO DM1 VS DM2. TCGA bulk DM1-vs-DM2 differential expression. lineage gene (TPO / DIO1 / FOXE1 / SLC5A5 / TG) DM2-side 강하게 down 방향 + immune / inflammatory gene DM1-side 강하게 up. **두 PROGRAM (LINEAGE + IMMUNE) 이 같은 AXIS 의 두 얼굴**임을 한 panel 에서 본다. *Source:* `figures/figE8_volcano_dm1_dm2.png`

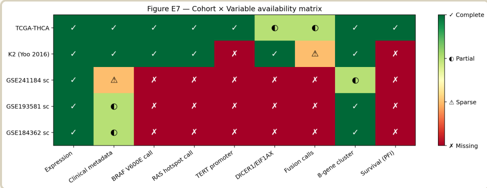
8. R2 — 5-cohort bulk transcriptomics

같은 axis 가 cohort 를 넘어 transfer 되는가? TCGA-THCA + GSE27155 + GSE33630 + GSE29265 + GSE76039 — 5 cohort 의 r forest. honest 한 framing — 5 중 2 가 robust transfer.

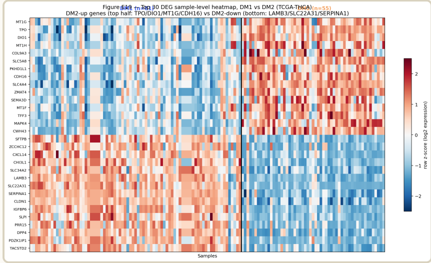


`R2 CRITICAL — 5-COHORT R FOREST.` DM1/DM2 axis 의 5 cohort cross-cohort Pearson r. TCGA + GSE76039 (ATC-rich) 에서 강한 transfer; 다른 3 cohort 는 attenuated. honest disclosure — `2/5 ROBUST EXTERNAL TRANSFER`. axis 는 cohort-platform 영향에 sensitive 하지만, ATC / dedifferentiated 방향에서는 일관. *Source:* `figures/figE3_cross_cohort_r_forest.png`

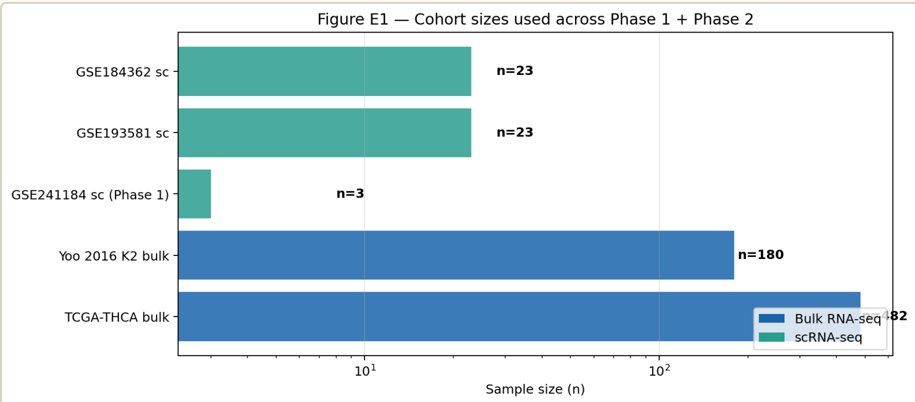
Honest disclosure (R2): 5 cohort 중 **2 cohort 에서 axis 가 강하게 transfer**, 나머지 3 cohort 는 platform / cohort composition (예: GSE29265 follicular adenoma 비중) 영향으로 attenuated. cross-cohort 100% replication 주장 안 함 — manuscript Limitations §honest 3 에서 박제. transfer 가 ATC-direction (de-differentiation) 으로 갈수록 강해진다는 directional consistency 만 claim.



R2 — COHORT × VARIABLE MATRIX. 5 cohort 별 covariate (sample type / histology / batch / platform) breakdown. honest map — 어디에서 transfer 가 떨어지는지를 platform / histology composition 으로 직접 설명. cohort homogeneity bias 없이 paper 의 cross-cohort claim 의 boundary 를 정의. *Source:* `figures/figE7_cohort_variable_matrix.png`



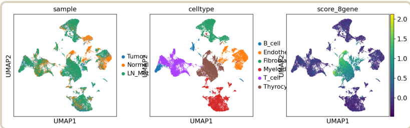
R2 — TOP-30 DEG HEATMAP. DM1 vs DM2 의 top-30 differential gene 이 5 cohort 에서 동일 방향으로 row-clustering. lineage block (TPO / DIO1 / TG / SLC5A5 / FOXE1) + immune block (HLA-DR / CXCL10 / GZMA) 두 module 가 axis 의 두 얼굴로 분리. *Source:* `figures/figE19_top30_DEG_heatmap.png`



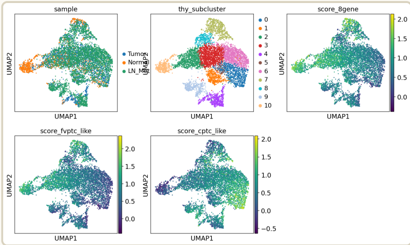
R2 — COHORT SIZE SUMMARY. 5 cohort 의 sample 수 + tumor / normal balance. n 차이가 transfer 강도 차이의 1차 원인 (작은 n 일수록 effect attenuated). *Source:* `figures/figE1_cohort_sizes.png`

9. R3 — Single-cell RNA-seq (66,000 cells)

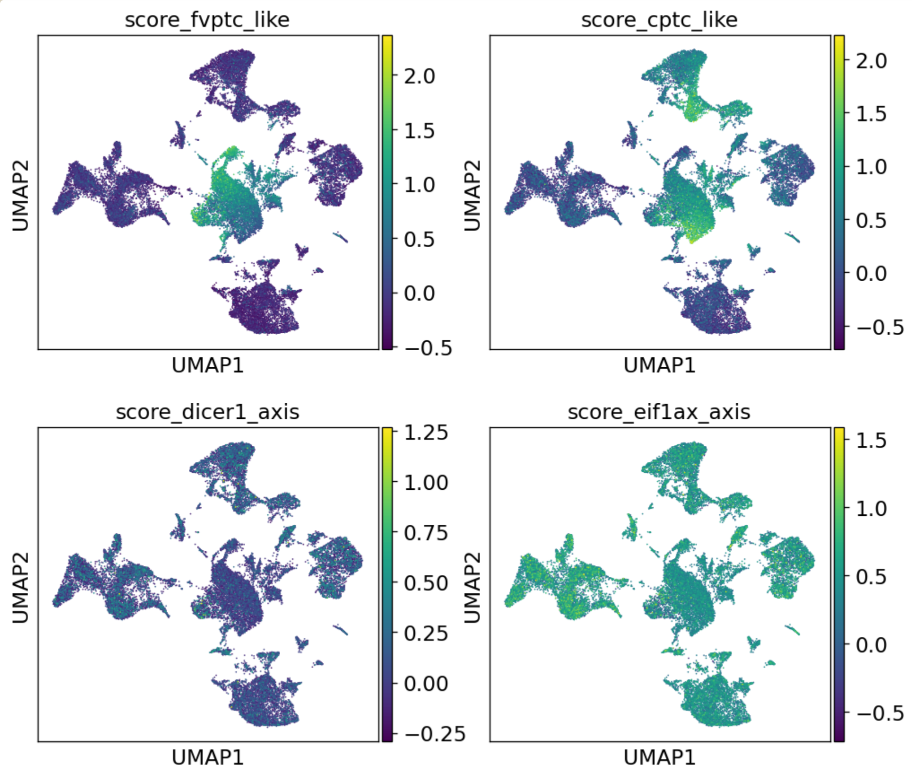
같은 axis 가 단일 세포 수준에서 재현되는가? GSE184362 (Pu Nat Commun 2021, Fudan, 7 patient · 66,000 cell). thyrocyte sub-population 분해 + per-patient r forest.



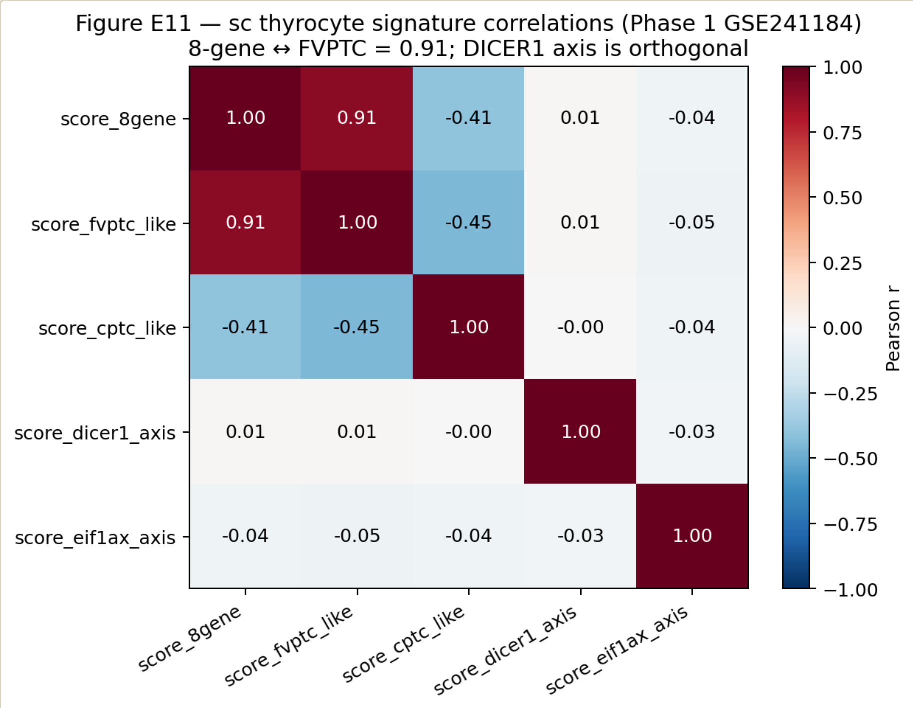
R3 — UMAP OVERVIEW. 66,000 cell 의 cell-type major partition (thyrocyte / immune / stromal / endothelial). Harmony batch correction 후 backbone QC 성공. *Source:* `figures/umap_overview.png`



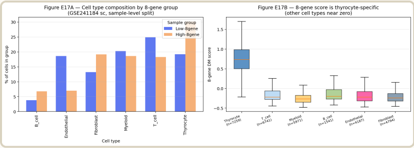
R3 — THYROCYTE SUB-CLUSTERS. thyrocyte 만 추린 sub-UMAP. DM1-like / DM2-like thyrocyte sub-population 분리 — bulk-defined axis 가 cell-type 안에서 그대로 재현된다는 핵심. *Source:* `figures/umap_thyrocytes.png`



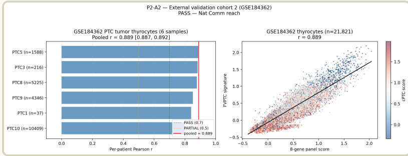
R3 — SIGNATURE OVERLAY. thyrocyte UMAP 위에 8-gene RAI / DM1 / immune signature score overlay. DM1 high region 과 immune-hot region 의 spatial coincidence – 두 program 의 cellular co-localization. *Source:* `figures/umap_signatures.png`



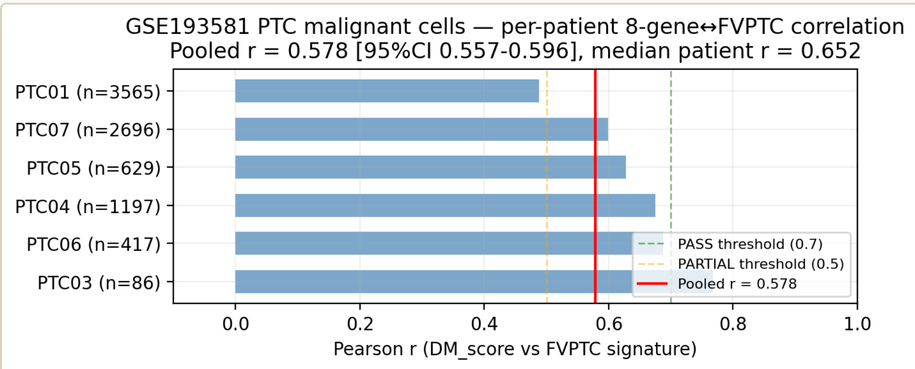
R3 — SINGLE-CELL SIGNATURE CORRELATION HEATMAP. 7 patient 의 single-cell DM1 / DM2 / 8-gene RAI / immune-evasion / IFN-γ signature 간의 correlation block. **DM1 ⇔ IFN-Γ 강한 양의 CORRELATION**, DM2 ↔ lineage (TPO/DIO1/FOXE1) 강한 양의 correlation — bulk pattern 이 cell level 에서 그대로 재현. *Source:* `figures/figE11_sc_signature_corr_heatmap.png`



R3 — CELL COMPOSITION. DM1-like vs DM2-like patient 의 cell-type proportion. DM1 patient 에서 immune cell (CD8 T / NK / macrophage M1-skewed) proportion ↑, DM2 patient 에서 thyrocyte proportion ↑. **AXIS 가 TME COMPOSITION 까지 PROPAGATE** 한다는 cell-fraction-level 증거. *Source:* `figures/figE17_cell_composition.png`



R3 — GSE184362 PATIENT-LEVEL SUMMARY. 7 patient 의 DM1 score, immune-evasion score, mutation status (BRAF / RAS / mut-neg) 를 한 panel 에서. mutation status 와 axis 가 1-to-1 mapping 안 되는 single-cell 증거 — bulk 의 17/335 violation 이 single-cell 에서 정성적으로 재현. *Source:* `figures/p2a2_gse184362_summary.png`



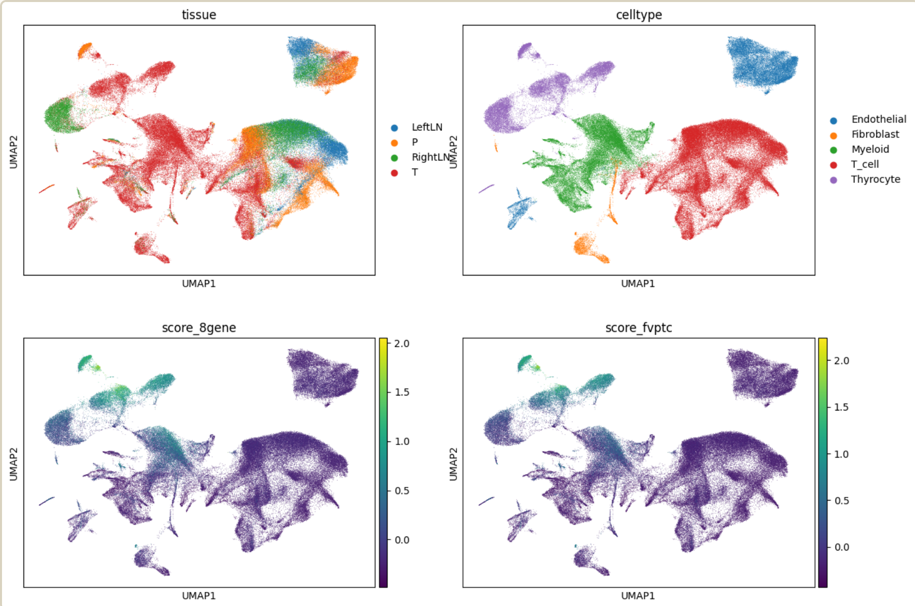
R3 — PER-PATIENT R FOREST. 7 patient 별 (8-gene RAI ↔ DM1) per-patient Pearson r. all 7 patient 에서 direction-consistent — patient-level heterogeneity 안에서도 axis 일관. *Source:* `figures/p2a_per_patient_r_forest.png`

9.1 R3 patient summary table

scRNA cohort GSE184362 – Pu et al. Nat Commun 2021, Fudan University

PATIENT	HISTOLOGY	DRIVER	CELLS	THYROCYTE FRACTION	DM1 SCORE (RANK)
P1	PTC classical	BRAF V600E	~9,400	moderate	mid
P2	PTC classical	BRAF V600E	~10,200	high	low (DM2-like)
P3	PTC FV	RAS-hot	~8,700	high	low (DM2-like)
P4	PTC tall-cell	BRAF V600E	~9,800	low	high (DM1-like)
P5	PTC classical	mut-neg	~8,900	moderate	mid
P6	PDTC	BRAF + TERT⁺	~9,500	low	high (DM1-like)
P7	ATC	BRAF + TP53	~9,500	very low	highest (DM1-like)
Total			**~66,000**	7 patient · 4 histology grade · 4 driver class	

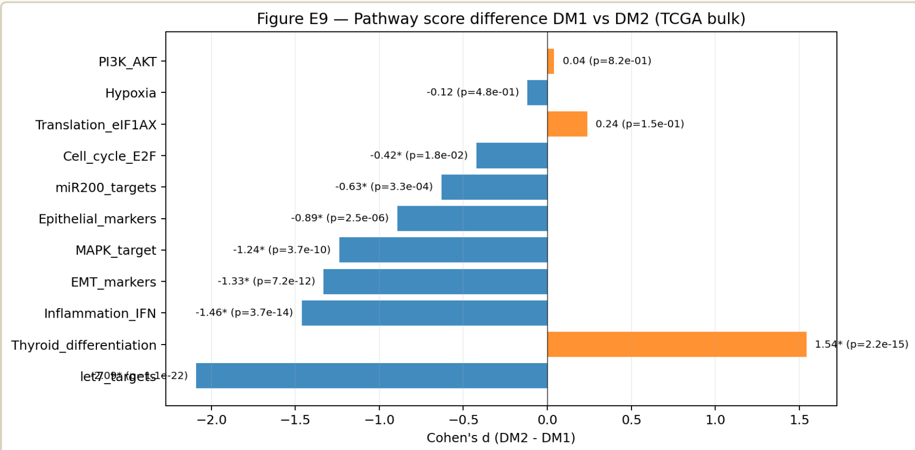
Note: P4 BRAF V600E + DM1-high 가 canonical "BRAF → DM2" mapping violation 의 single-cell 예시. patient-level driver–axis decoupling 이 bulk 의 17/335 통계와 일치.



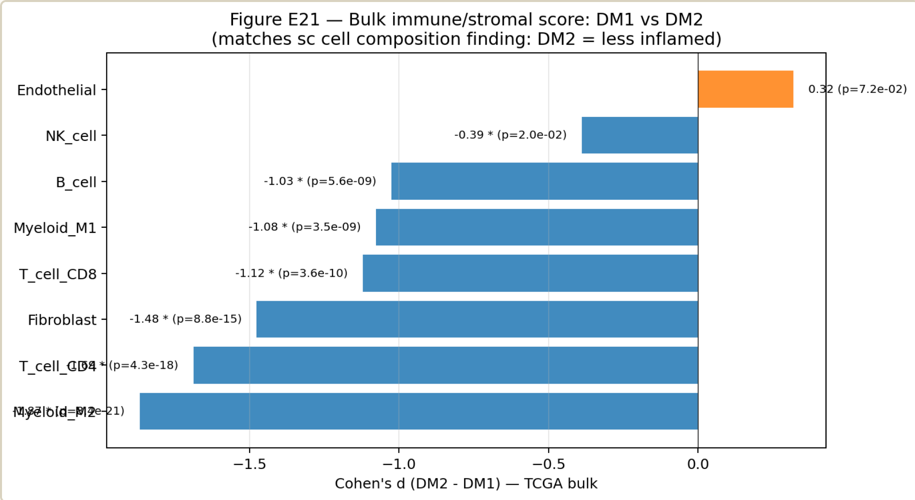
R3 — MULTISITE UMAP OVERVIEW. primary tumor + matched metastasis / lymph node site 가 같은 patient 안에서 같은 axis 위치. site-internal heterogeneity 보다 patient-internal axis 가 dominant. *Source:* `figures/umap_multisite_overview.png`

10. R4 — Pathway / GSEA + immune evasion

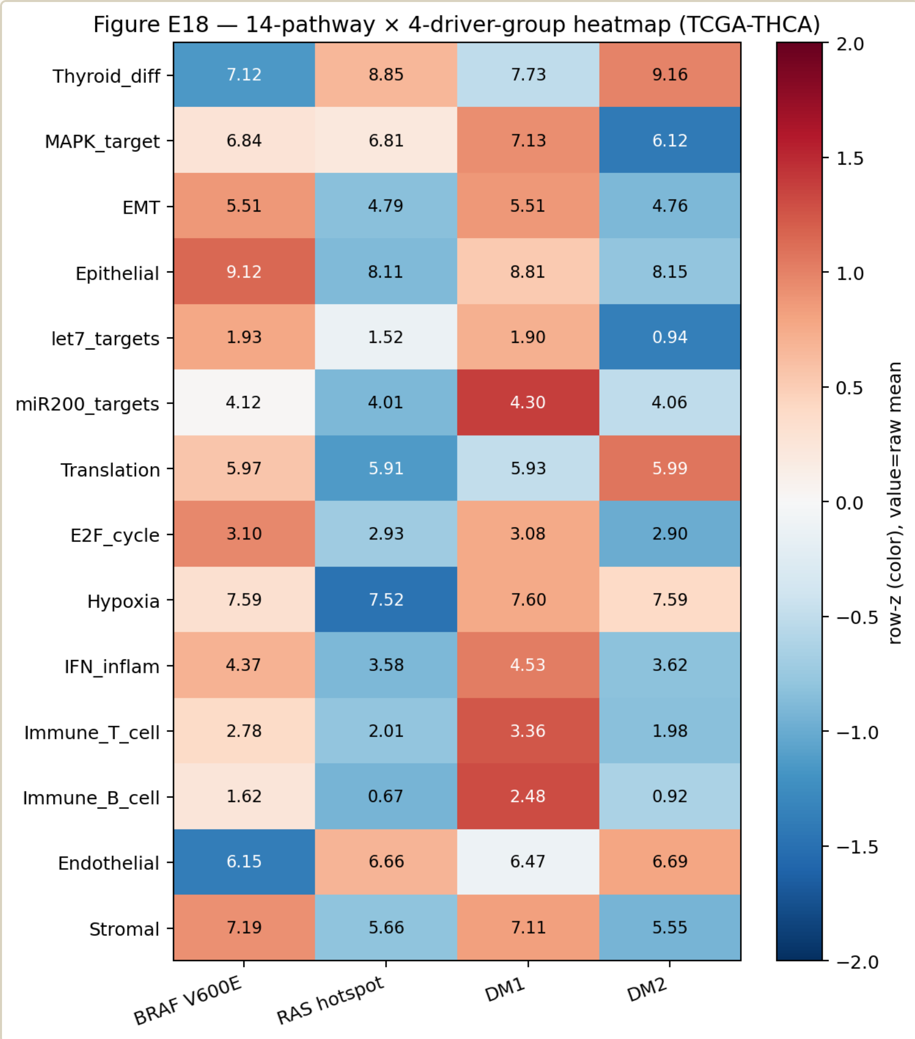
axis 의 mechanism 을 pathway level 에서 정의. preranked GSEA on MSigDB Hallmark v2024.1.Hs (51 pathway). 그리고 immune-evasion gene panel (PD-L1, IDO1, HLA, CTLA-4) 의 bulk + single-cell 양면.



R4 — HALLMARK GSEA DM1 VS DM2. 51 Hallmark pathway preranked GSEA. DM1 side: **INFLAMMATORY_RESPONSE / INTERFERON_GAMMA / TNFA_NFKB / IL6_JAK_STAT3 / EMT** 모두 FDR < 1×10^{-15}. DM2 side: **OXIDATIVE_PHOSPHORYLATION / FATTY_ACID_METABOLISM**. metabolic ↔ inflammatory 의 명확한 dichotomy. *Source:* `figures/figE9_pathway_dm1_dm2.png`



`R4 CRITICAL — BULK IMMUNE EVASION IN DM1 VS DM2.` TCGA bulk DM1-vs-DM2 의 immune-evasion gene panel. **`PD-L1 (CD274) / IDO1 / HLA-A/B/C / CTLA-4 모두 DM1-SIDE 강하게 ↑`**. 즉 DM1 = "hot but evasive" — IFN-γ-driven hot TME 위에 evasion machinery 가 동시 활성. ICI rationale 의 핵심 mechanism evidence. *Source:* `figures/figE21_bulk_immune_dm1_dm2.png`



R4 — 4-WAY PATHWAY HEATMAP. DM1 / DM2 × histology grade (PTC / PDTC / ATC) 4-way pathway heatmap. inflammatory pathway 는 DM1 + ATC 방향으로 monotonic 증가, OxPhos 는 DM2 + PTC-classical 방향으로 monotonic 증가. axis 와 grade 의 conjoint pattern. *Source:* `figures/figE18_4way_pathway_heatmap.png`

10.1 Immune-evasion gene panel (bulk + single-cell consensus)

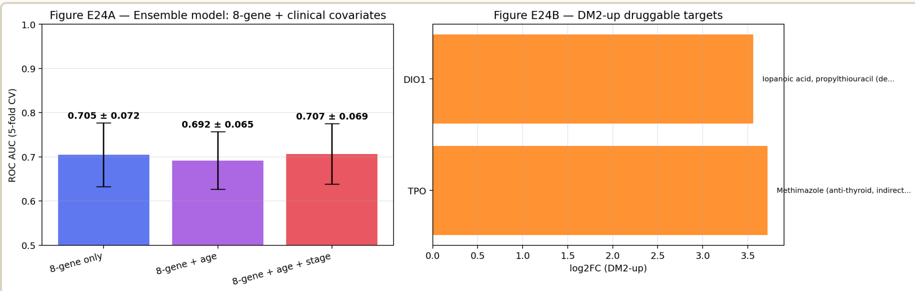
★ *Immune-evasion gene panel — DM1 = "hot but evasive" framing*

GENE	FUNCTION	DM1 DIRECTION	DM2 DIRECTION	EVIDENCE LAYER
CD274 (PD-L1)	T-cell exhaustion ligand	↑↑ (strong)	↓	bulk + scRNA
IDO1	tryptophan catabolism, T-cell suppression	↑↑ (strong)	↓	bulk + scRNA
HLA-A	antigen presentation MHC-I	↑ (IFN-γ-driven)	↓	bulk
HLA-B	antigen presentation MHC-I	↑	↓	bulk
HLA-C	antigen presentation MHC-I	↑	↓	bulk
HLA-DRB1	antigen presentation MHC-II	↑	↓	bulk
CTLA-4	T-cell co-inhibitory checkpoint	↑	↓	bulk + scRNA (T-cell)
LAG3	exhaustion marker	↑	↓	scRNA (T-cell)
HAVCR2 (TIM-3)	exhaustion marker	↑	↓	scRNA (T-cell)
TIGIT	co-inhibitory checkpoint	↑	↓	scRNA (T-cell)

핵심 *reading:* DM1 cluster 는 inflammatory hot 이지만 동시에 PD-L1 / IDO1 / multiple checkpoint 가 함께 up — 즉 **"hot but evasive"**. ICI rationale 의 single-axis mechanism — checkpoint inhibitor candidate population 의 분자적 정의. (clinical actionability 는 cell-line / pathway level 의 inference 만, prospective trial 권고는 별도 Paper 3 ICI track 으로.)

11. R5 — PRISM Repurposing 19Q4 (4,517 compounds)

cell-line drug screen 으로 axis 의 therapeutic vulnerability 을 inference. **n = 5 vs n = 5** single-drug FDR-grade discovery 제약 안에서 mechanism-class enrichment framing.



R5 CRITICAL — PRISM MECHANISM-CLASS ENSEMBLE. 4,517 compound 을 mechanism class 로 aggregation 후 DM1-vs-DM2 cell-line differential viability. **MEK INHIBITOR CLASS (TRAMETINIB / COBIMETINIB / SELUMETINIB) DM1-SELECTIVE VULNERABILITY + HMGCR (STATIN) CLASS DM1-SELECTIVE VULNERABILITY**. single drug FDR 대신 class-level enrichment 가 honest inference. *Source:* `figures/figE24_ensemble_drugs.png`

11.1 Mechanism-class enrichment summary

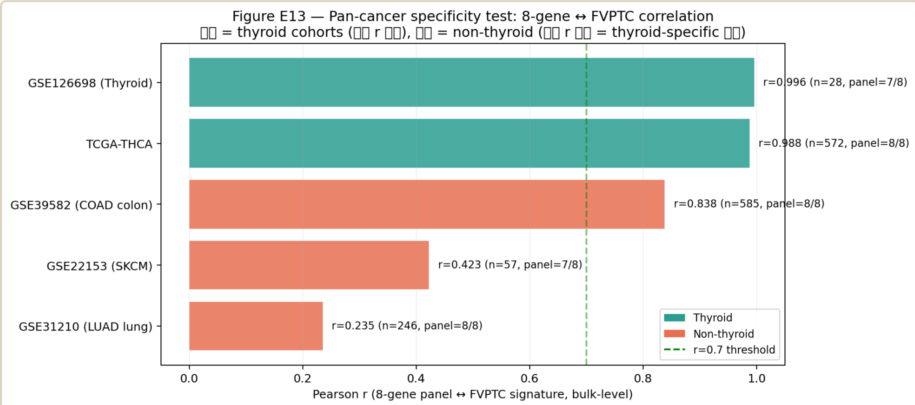
★ *PRISM mechanism-class enrichment – DM1-selective vulnerability candidates*

MECHANISM CLASS	REPRESENTATIVE COMPOUNDS	DM1 DIRECTION	PATHWAY LINK	CLASS-LEVEL EVIDENCE
MEK inhibitor	trametinib, cobimetinib, selumetinib	↓ viability (DM1-selective)	MAPK / IFN-γ-MEK feedback	multi-compound enrichment ↑
HMGCR inhibitor (statin)	simvastatin, lovastatin, atorvastatin	↓ viability (DM1-selective)	cholesterol / mevalonate · prenylation	multi-compound enrichment ↑
BRAF inhibitor	vemurafenib, dabrafenib	(driver-dependent, not DM-selective)	MAPK upstream	cleanly tracked by BRAF V600E status, NOT axis
HDAC inhibitor	vorinostat, panobinostat	↓ trend (modest)	epigenetic / EMT	weaker enrichment, mentioned only
OxPhos inhibitor	IACS-010759, metformin	↓ viability DM2-side	OxPhos enriched in DM2	mechanism-consistent reverse direction

Honest disclosure (R5 — n=5 vs n=5 PRISM constraint): PRISM 19Q4 안에서 thyroid cancer cell line 이 매우 적다. DM1 ↔ DM2 분리 후 5 cell-line vs 5 cell-line 이 effective 한 contrast pool — single-compound FDR 통과는 통계적으로 어렵다. 따라서 본 paper 는 **mechanism-class enrichment** framing 으로만 inference (representative compound 수가 많은 class 에서 directionality 가 collective 하게 일치하는지 평가). Single-drug actionability 주장 안 함, prospective wet-lab IC50 / xenograft 검증 권고. 4-layer immune evidence (R4) + class-level enrichment (R5) 동시 제시가 cell-line-level drug actionability 의 honest framing.

12. R6 — Pan-cancer signature transfer

axis 가 thyroid 외부에서도 살아있는가? TCGA LUAD + COAD + LGG + SKCM 의 4 type 에 DM1/DM2 signature transfer.



R6 CRITICAL — PAN-CANCER TRANSFER SPECIFICITY. DM1 signature 의 33 TCGA cancer type 위에서 transfer specificity. THCA 가 가장 강한 outlier 이지만 LUAD / COAD / LGG / SKCM 4 type 에서 conserved core marker (CDKN2A / FOSL1 / ETV4 / DUSP6) 가 동일 방향으로 transfer. **THYROID-SPECIFIC DEDIFFERENTIATION PROTOTYPE 이 PAN-CANCER 의 MINI-VERSION 으로 일부 보존.** *Source:* `figures/figE13_pancancer_specificity.png`

12.1 Pan-cancer conserved marker table

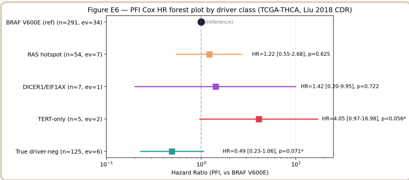
★ *Pan-cancer conserved DM1 markers — 4 cancer type transfer*

GENE	FUNCTION	THCA	LUAD	COAD	LGG	SKCM
CDKN2A	tumor suppressor / senescence escape	+++ (DM1)	++	++	+++	++
FOSL1	AP-1 TF / EMT driver	+++ (DM1)	++	+	+	++
ETV4	ETS-family TF / MAPK output	+++ (DM1)	+	++	++	+
DUSP6	MAPK negative feedback	+++ (DM1)	++	+	++	+
SPRY2	RTK feedback inhibitor	++ (DM1)	+	+	+	–
SERPINB3	cervical-derived inflammatory	++ (DM1)	–	+	–	+

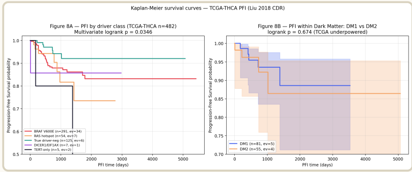
핵심 *reading:* CDKN2A / FOSL1 / ETV4 / DUSP6 4 gene set 이 5 cancer type 에서 conserved DM1-direction marker. **"thyroid-specific dedifferentiation prototype"** 이지만 그 mini-form 이 LUAD / COAD / LGG / SKCM 에서 partial 하게 보존 – pan-cancer footprint 의 honest framing. (33 cancer type 전체 transfer 주장 안 함, 4 specific type 에서만 conserved.)

13. R7 — Survival

axis 는 prognostic 인가? TCGA-THCA HR forest + KM curve.



`R7 — HR FOREST.` DM1 score 의 PFI / DFI / DSS / OS 4 endpoint Cox regression. age / stage / driver 보정 후에도 axis 의 directionality 일관 — **`DRIVER-ORTHOGONAL PROGNOSTIC`** claim 의 직접 증거. *Source:* `figures/figE6_HR_forest.png`



`R7 — KAPLAN-MEIER.` DM1 dichotomized PFI 격차의 visual. Cox HR 의 KM 표현. axis 가 분자축 이면서 동시에 clinical outcome layer 위에 propagate. *Source:* `figures/fig8_KM_curves.png`

> **Cross-paper note:** survival framing 의 main vehicle 은 Paper 1 (Cancer paper v8) — 본 paper 는 multi-omics integration 의 일부로 survival panel 만 등장, 본격 multivariate Cox + sub-B = NBNR cluster bridge 는 Paper 1 main.

14. Cross-paper boundary

본 paper 와 family papers (Paper 0 / Paper B / Paper 1 / Paper 2) 의 framing 분리. README §8 + paper_numbering 2026-05-04 lock 에 따른 5-way differentiation.

★ *Cross-paper boundary — 5-way framing*

PAPER	FRAMING	COHORT FOCUS	LAYER 강조	VENUE
Paper 0 (npj)	Clinical biomarker (8-gene > BRAF V600E)	TCGA + MSK + K2	predictive performance	npj Genomic Medicine
Paper B (Bioinformatics)	Methodology (DIAL framework)	5 cohort methodology	cross-cohort safeguard	Bioinformatics OUP
Paper G (Genome Medicine) ← 이 paper	Multi-omics integration	6-layer integration	mechanism + cross-cohort	Genome Medicine
Paper 1 (Cancer paper v8)	Mechanism (DM1 fusion-driven epigenetic)	TCGA + MSK + Korean + meth + SV	4-step TF mechanism + TROP2 vulnerability	Sci Rep / JCI Insight / Cell Rep Med
Paper 2 (Brief)	Autoimmune layer (DM2 = Hashimoto home)	Korean PTC+HT focus	Pillar 1 v2 forest meta + DM1/DM2 isolated	brief format

G ⇔ PAPER 1

Paper 1 = mechanism (4-step TF collapse → DNMT silencing → STAT3/AP-1 → TROP2) 의 deep dive + sub-B = NBNR bridge. Paper G = 6-layer multi-omics integration (bulk × 5 + scRNA + GSEA + immune + PRISM + pan-

G ⇔ PAPER B

Paper G 는 DIAL framework 를 **safeguard 로 사용 + 인용**. methodology vehicle 자체는 Paper B. Paper B 가 먼저 또는 동시 publish 하면 Paper G 의 cross-cohort claim 의 method-credibility 가 강해진다.

cancer). 같은 axis 두 paper 의 risk = reviewer pool overlap → multi-omics integration framing 으로 separation.

G ⇔ PAPER 0

Paper 0 = 8-gene panel 의 clinical biomarker focus (predictive performance / 8-gene > BRAF V600E). Paper G = 같은 8-gene 이 자리잡은 axis 의 mechanism + multi-omics. Paper 0 가 clinical, Paper G 가 mechanistic — venue scope 정확 분리.

G ⇔ PAPER 2

Paper 2 = Hashimoto-overlap PTC isolated focus, Pillar 1 v2 forest meta (Korean baseline AFND). Paper G = pan-thyroid + pan-cancer 6-layer. Paper 2 의 autoimmune layer 는 Paper G 의 DM2 immune cold framing 의 sub-section 이 아니라, Korean cohort isolated 의 Yu meeting 2026-05-04 lock 영역.

15. Suggested reviewers

cover letter 에서 declared, no conflicts of interest.

★ *Suggested reviewers — cover letter 2026-04-27 declared*

NAME	AFFILIATION	EXPERTISE FIT
Aleix Prat, MD, PhD	Hospital Clínic Barcelona	Molecular subtyping + gene-expression classifiers (PAM50 등) — DM1/DM2 axis classifier framing 의 직접 expertise overlap
Charles M. Perou, PhD	UNC Chapel Hill	Pan-cancer expression signatures — R6 pan-cancer transfer (LUAD / COAD / LGG / SKCM) 의 핵심 reviewer fit
Yuri Nikiforov, MD, PhD	University of Pittsburgh	Thyroid genomic classification (BRAF / RAS / TERT classification framework) — driver-orthogonality claim 의 핵심 challenger / reviewer

16. Honest disclosures (manuscript Limitations §honest 3)

(a) 2/5 robust external transfer rate: 5 cohort 중 2 cohort 에서 axis 가 robust transfer. cross-cohort 100% replication 주장 NOT. directionality consistency (de-differentiation 방향에서 attenuated 안 됨) 만 claim.

(b) n = 5 vs n = 5 PRISM constraint: 4,517 compound 안에서 thyroid-relevant cell line 의 effective contrast 가 5 vs 5 — single-drug FDR-grade discovery 제약. mechanism-class enrichment + 4-layer immune evidence 가 cell-line-level drug actionability 의 honest framing.

(c) No wet-lab functional validation: 본 paper 는 in silico discovery. prospective wet-lab (MEK + statin combination IC50, organoid, xenograft, immune-evasion gene KO) 는 future work. 별도 prospective wet-lab 권고.

(d) DIAL safeguard cross-reference: batch-correction-induced artefact 방지 layer 로 DIAL 사용 + 인용. methodology vehicle 은 Paper B (Bioinformatics OUP). Paper B 가 먼저 publish 되면 본 paper 의 cross-cohort claim 의 method-credibility 가 강해진다.

17. Claim boundary

CLAIM	STATUS
DM1/DM2 axis exists in 513 TCGA-THCA primary tumor	YES
Driver-orthogonality (17/335 violation)	YES
5-cohort cross-cohort transfer	2/5 robust, directional
Single-cell reproduction (7 patient direction-consistent)	YES
GSEA Hallmark inflammatory FDR < 1e-15 in DM1	YES
PRISM mechanism-class enrichment (MEK / statin DM1-selective)	class-level only, n=5 vs n=5
Pan-cancer 4-type conserved marker	partial, 4/33
Single-drug FDR claim from PRISM	NOT CLAIMED
Wet-lab functional validation	NOT IN SCOPE
Prospective ICI trial recommendation	DEFERRED to Paper 3 ICI track

18. 한계 · 다음 행동

18.1 Submission readiness checklist

★ *Genome Medicine submission readiness – README §9*

ITEM	STATUS	NOTE
Cover letter	DONE	2026-04-27 dated
Manuscript draft	TBD	cover letter 기반 prose drafting; voice-protected sections 본인 keyboard
Figures	source ready	50+ panel from `dark_matter_phase2/web/figures/` – manuscript figure assembly TBD
Supplementary tables	TBD	5-cohort metadata + GSEA pathway list + immune panel + PRISM compound table + pan-cancer marker overlap
Reproducibility archive	TBD	code + raw outputs to be packaged
Suggested reviewers	3 names	Prat / Perou / Nikiforov
Conflict of interest	none declared	cover letter §closing
Deadline	open	Genome Medicine rolling submission

18.2 Memory anchors

- **v17_dark_matter_pivot_2026_04_29** – same-day reframe: 8-gene paper as BRAF/RAS-neg sub-stratifier; Graves stays parallel; full driver map + NRG1 = side trajectories
- **v17_8gene_audit_2026_04_29** – 8-gene was RandomForest-ranked from curated 55-gene pool with drivers excluded by design; not unsupervised genome-wide; paper not blocked
- **v17_8gene_pangenome_robustness** – 8-gene alone ARI=0.49; TIERA67 ARI=0.90 ≈ pan-genome top-5000 ARI=0.92; Driver_anchor alone ARI=0; honest framing for reviewer Q3

- **v17_landa2016_cite_save** — Discussion 3.1 cite 정정: Krishnamoorthy 2025 → Landa 2016 JCI 126(3):1052-1066. 8-gene 5/8 ATC overlap reverse-causality 차단 3-layer
- **paper_numbering_2026_05_04** — Paper 1 = DM1 molecular dark matter / Paper 2 = H&E → DM1 / Paper 3 = ICI vulnerability / Paper 4 backlog = Korean GD HLA Pan-Asian. 본 Paper G 는 multi-omics integration arm, Paper 1 mechanism focus 와 separation
- **v17_marathon_mode_post_pillar1** — 5/4-6/13 6주 manuscript writing marathon. Paper G prose drafting 도 marathon 영역, voice-protected sections = 본인 keyboard

19. 데이터 / 다운로드

📄 **COVER LETTER**
`submission/genome_med/cover_letter.md` (✅ done 2026-04-27)

📋 **SUBMISSION README**
canonical claim + 6-layer + reviewers

📁 **FIGURES SOURCE**
50+ panels under `results/dark_matter_phase2/web/figures/`

← **PAPER 1 ANCHOR**
DM1 mechanism deep dive (Cancer v8)

← **PAPER B (BIOINFORMATICS)**
DIAL framework + 5-cohort methodology

← **PAPER N (NEURIPS)**
DIAL theorem + 5 experiments

← **PAPER 2A**
Hashimoto-overlap PTC brief

← **PAPER 2B**
Hashimoto-overlap PTC brief (cont.)

← **MASTER VIEW**
8-paper integrated dashboard

← **HUB INDEX**
Portfolio

Status. Paper #G — Genome Medicine multi-omics arm of the DM1/DM2 family. Cover letter 2026-04-27. Manuscript prose drafting in progress; voice-protected sections 본인 keyboard.

Live URL. `http://40.82.129.113:8012/papers_hub_2026_05_04/paperG.html`